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Magnetostratigraphic investigations of the Middle Triassic Badong Formation in South China

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SUMMARY

Magnetostratigraphic sampling of the Middle Triassic Badong Formation in South China was conducted at three sections. A dual-polarity characteristic remanent magnetization (ChRM) resolved from most samples by thermal demagnetization is shown to have been acquired prior to folding. The primary nature of the ChRM is corroborated by the discovery of the same magnetic polarity at equivalent stratigraphic levels in more than one section. The relative sample VGP (virtual geomagnetic pole) latitudes define nine magnetozones for the three major constituent members of the formation. Comparison with the Mid-Triassic magnetic polarity sequence observed from the western Tethyan region appears to indicate that the bulk of the Badong Formation is Anisian in age and that Ladinian sediments are largely missing. This agrees with palaeontological and stratigraphic evidence in the region and supports the view that Ladinian regression is a major event in the geological evolution of South China, which may signal the onset of amalgamation of the Yangtze Block (YB) with the North China Block (NCB).

Key words: Badong Formation, magnetostratigraphy, South China, Triassic.

INTRODUCTION

Magnetostratigraphy of terrestrial or marine rocks of Triassic age has been the subject of a number of investigations in recent years (e.g. Steiner *et al.* 1989; Kent *et al.* 1995; Molina-Garza *et al.* 1996; Muttoni *et al.* 1996, 1997; Gallet *et al.* 1998), reflecting the increased efforts to extend the magnetic polarity timescale (MPTS) beyond the Jurassic Period. In comparison, Middle Triassic magnetic stratigraphy is less well established (Opdyke & Channell 1996). To amend this deficit, we conducted a magnetostratigraphic study of the Middle Triassic Badong Formation in South China. The advantage of the Badong Formation for this purpose is that it contains palaeomagnetically favourable lithology (redbeds) and is well exposed along several long sections in the study area. In this paper, we will first present the magnetostratigraphic data obtained, and then correlate the data with the magnetic polarity sequence reported from the western Tethyan region and discuss the geological implications.

LITHOLOGY AND FOSSIL RECORDS OF THE BADONG FORMATION

The Badong Formation crops out in the border area between Hubei, Sichuan, Guizhou and Hunan provinces (Fig. 1). It conformably overlies Lower Triassic shallow marine carbonates, and, in most places, underlies Upper Triassic terrestrial coal series. The age of the underlying and overlying rocks is well controlled by fossils. The type section of the Badong Formation in western Hubei is divisible into five members (Zhang *et al.* 1987): member 1 is 80 m thick and consists of grey calcareous shale interbedded with thin-bedded limestone and dolomitic limestone; Member 2 is 419 m in thickness and is composed of purplish-red calcareous siltstone and mudstone; Member 3 is characterized by thin- to medium-bedded limestone with a thickness of 272 m; Member 4 is also composed of purplish-red calcareous siltstone and silty mudstone and is 356 m thick; and Member 5 is made up mainly of dark grey dolostone with a thickness of 25 m. Members 1, 3, and 5 are considered to be platform carbonate deposits, and Members 2 and 4 are regarded as tidal flat sediments (Zhang *et al.* 1987). This sequence represents the Badong Formation at most sections in the area, and the limestones of Member 3 are particularly useful correlation markers. Member 1 contains a rich bivalve fauna, represented by *Costatoria goldfussi mansuyi* and *Leptochondria illyrica*, which is commonly seen in the rock formations of Anisian age in southwestern China. Member 3 yields abundant marine fossils from which a bivalve assemblage (*Plagiostoma striatum–Placunopsis plana*), an ammonoid zone (*Progonoceratites*), and a conodont zone (*Neospathodus kockeli*) have been recognized (Zhang *et al.* 1987). These fossils indicate that Members 1 to 3 are Anisian in age. Although Members 4

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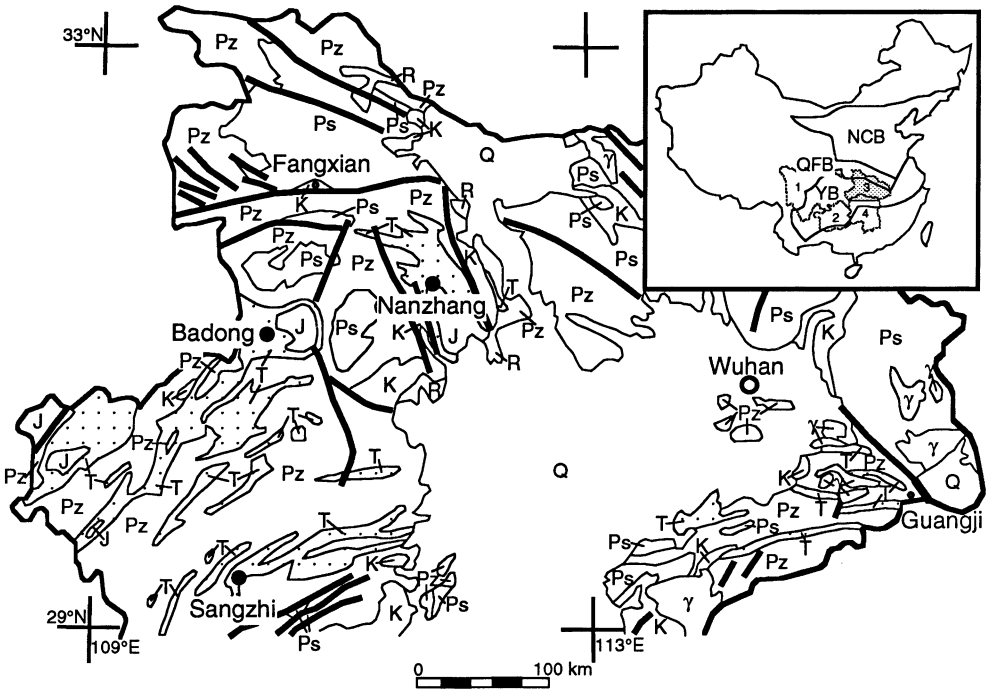


Figure 1. Geological sketch map of Hubei and the northern Hunan provinces. Solid circles denote the sampled sections of this study. Strata: Ps = pre-Sinian; Pz = Sinian-Permian; T = Triassic; J = Jurassic; K = Cretaceous; R = Tertiary; Q = Quaternary. Triassic outcrops are stippled. Granitic intrusions are indicated with γ. Heavy lines are major faults. In the inset, the stippled area is the area the map shows. Broken lines are the boundaries of Sichuan (1), Guizhou (2), Hubei (3), and Hunan (4) provinces. Abbreviations for the tectonic units: NCB = North China Block; YB = Yangtze Block; QFB = Qinling Fold Belt.

and 5 yield few diagnostic fossils, they have been considered Ladinian in age because they lie conformably above Member 3 and below the fossiliferous Late Triassic coal series (Zhang *et al.* 1987).

MAGNETOSTRATIGRAPHIC SAMPLING AND RESULTS

Palaeomagnetic samples were collected from three localities using a portable rock drill: Badong in western Hubei (31.0°N, 110.4°E), Sangzhi (29.4°N, 110.2°E) in northwestern Hunan, and Nanzhang (31.4°N, 111.7°E) in central Hubei (Fig. 1). Laboratory work was conducted at the Palaeomagnetic Laboratory of the University of Florida. All the samples were subjected to stepwise thermal demagnetization. Standard palaeomagnetic data analysis techniques (Fisher 1953; Zijderveld 1967; Kirschvink 1980) were utilized.

Badong, western Hubei

Sampling at this type locality was carried out at three subsections: Members 1 and 2 were sampled along a ravine (Lujiagou) near downtown Badong on the southern bank of the Yangtze river; Members 4 and 5 were sampled along a road (Baotahe) on the northern bank of the river across from downtown Badong (both these subsections are situated on the southern limb of a W-trending syncline); and Member 3 was sampled 2 km north of Badong along the road to Xingshan on the northern limb of the syncline because the limestones of Member 3 on the southern limb are mostly submerged under the river. The sampling interval varied from about 5 m in Member 2 to 10 m in Members 3 and 4. The top 19 m of Member 2 could not be sampled due to residential housing. A total of 137 core samples were collected from Members 1 (6), 2 (61), 3 (27), 4 (37), and 5 (6). Seven more sites, usually consisting of six cores, were drilled from Member 2 at the third subsection for a fold test.

The palaeomagnetic results derived from Member 2 have been reported elsewhere (Huang & Opdyke 1996). Briefly, the natural remanent magnetization (NRM) of most samples from Member 2 is composed of a low-temperature post-folding component (A) and a high-temperature pre-folding ChRM component which is of dual polarity (Figs 2a, 2b and 3; see also Figs 2 and 3 in Huang & Opdyke 1996). The sample VGP latitudes relative to the overall mean palaeomagnetic (north) pole define a dominant normal-polarity magnetozone with a short reversed-polarity magnetozone near the bottom of Member 2 (Fig. 8). A shorter reversed magnetozone at the top is also indicated by the results from two sites (n1 and n2) drilled in the topmost part of Member 2 on the northern limb of the syncline. Although site n1 did not yield well-defined ChRM directions due to weak NRM intensities, the high-temperature directions are apparently of reversed polarity (Fig. 2c). Three samples out of six from site n2 gave reversed ChRM directions (see Figs 2e in Huang & Opdyke 1996).

Unfortunately, limestone samples from Members 1, 3, and 5 did not yield any usable results. The NRM of most samples from Member 4 was dominated by the post-folding component A (Fig. 2d), and only 10 samples yielded the high-temperature dual-polarity ChRM directions (Figs 2e and f). The relative VGP latitudes of these samples define at least one reverse- and one normal-polarity magnetozone near the top of Member 4 (Fig. 8).

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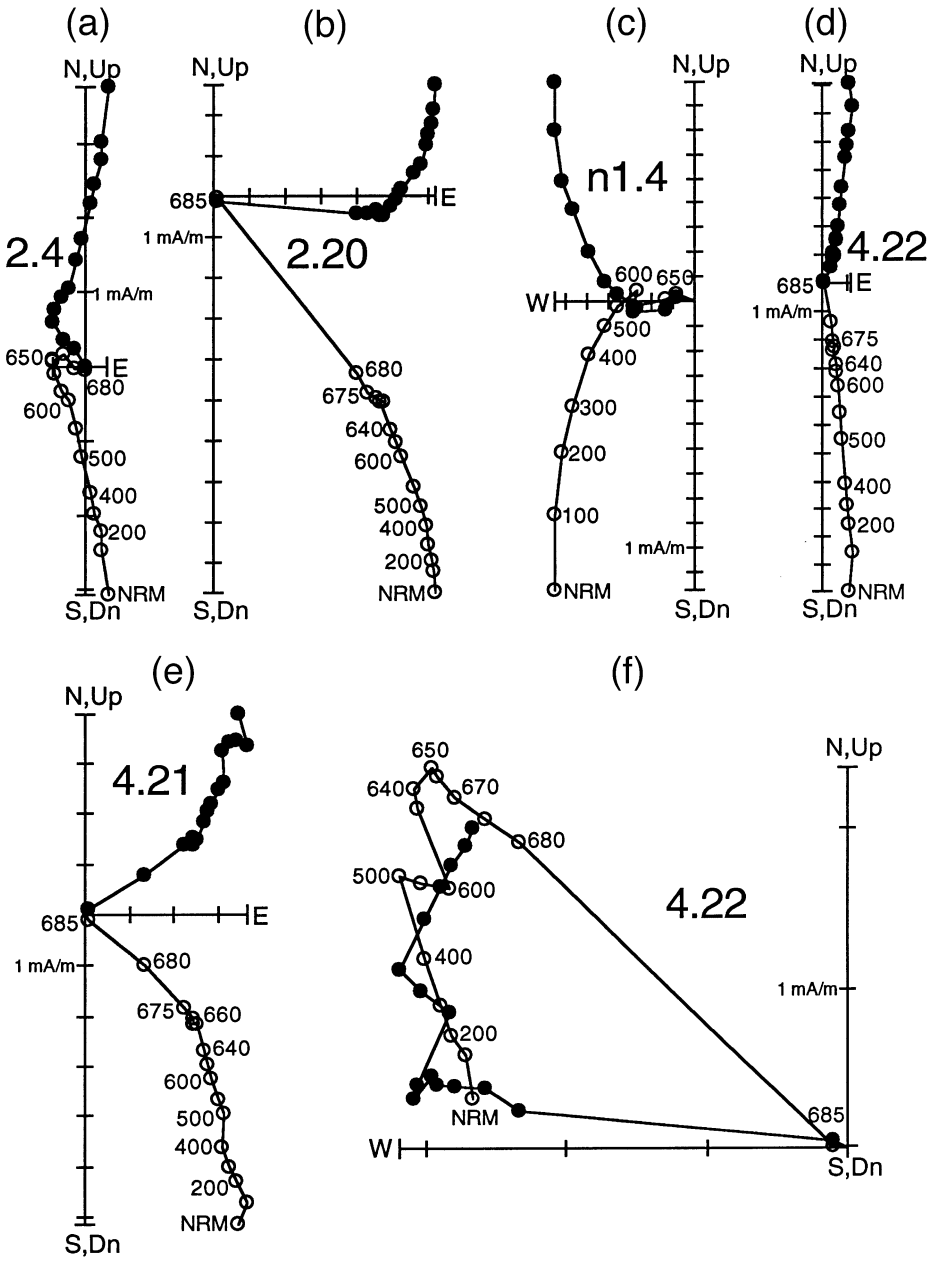


Figure 2. Representative orthogonal plots of thermal demagnetization (*in situ* coordinates) for the samples from Badong. (a), (b) and (c) are from Member 2, and (d), (e) and (f) from Member 4, (c) from the northern limb of the syncline. Solid (open) circles are plotted on horizontal (vertical) planes. Treatment levels are in degrees Celsius and are indicated along the demagnetization trajectory. The large numbers such as 2.4 in the plots are the sample numbers.

Sangzhi, northwestern Hunan

The Hongjiaguan section is located 9 km north of downtown Sangzhi and is situated on the northwestern limb of a NE-trending synclinorium. The total thickness of the Badong Formation at this section is nearly twice that at Badong (Hunan Bureau of Geology 1989). Magnetostratigraphic sampling at this section was only carried out on Member 3, because of its favourable lithology, namely thin-bedded argillaceous limestone intercalated with reddish silty marl and siltstone. The thickness of Member 3 at Hongjiaguan is 218 m, and 25 samples, including two from the base of Member 4, were collected.

20 sites were drilled from Member 2 from two limbs of the synclinorium, and these results are presented in Huang & Opdyke (1996). Thermal demagnetization resolved a dual-polarity, pre-folding ChRM for all the sites from Member 2 (Figs 4a and b; see also Figs 5 and 6 in Huang & Opdyke 1996). A similar dual-polarity ChRM was also isolated from 20 out of 25 samples from Members 3 and 4 (Figs 4c, 4d and 5); limestone samples once again failed to yield usable results. The first sample taken from the base of Member 3 is of reversed polarity, which appears to be continuous with the short reversed magnetozone detected at the top of Member 2 at Badong. The successive 11 samples above define

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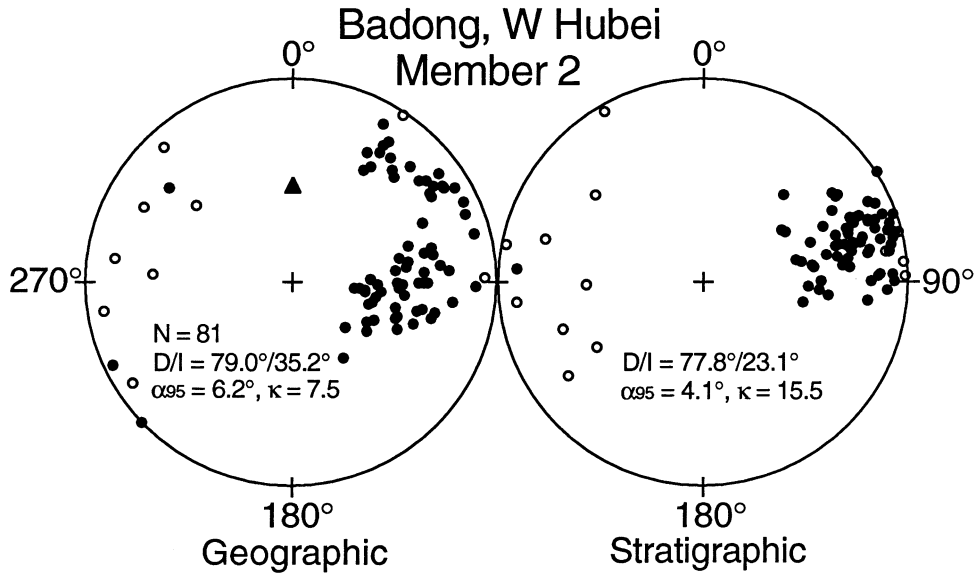


Figure 3. Equal-area projection of the ChRM directions of the samples from Member 2 from Badong. Solid (open) circles are projections on the lower (upper) hemisphere. The solid triangle denotes the dipole direction of the Earth's present-day magnetic field at the sampling locality.

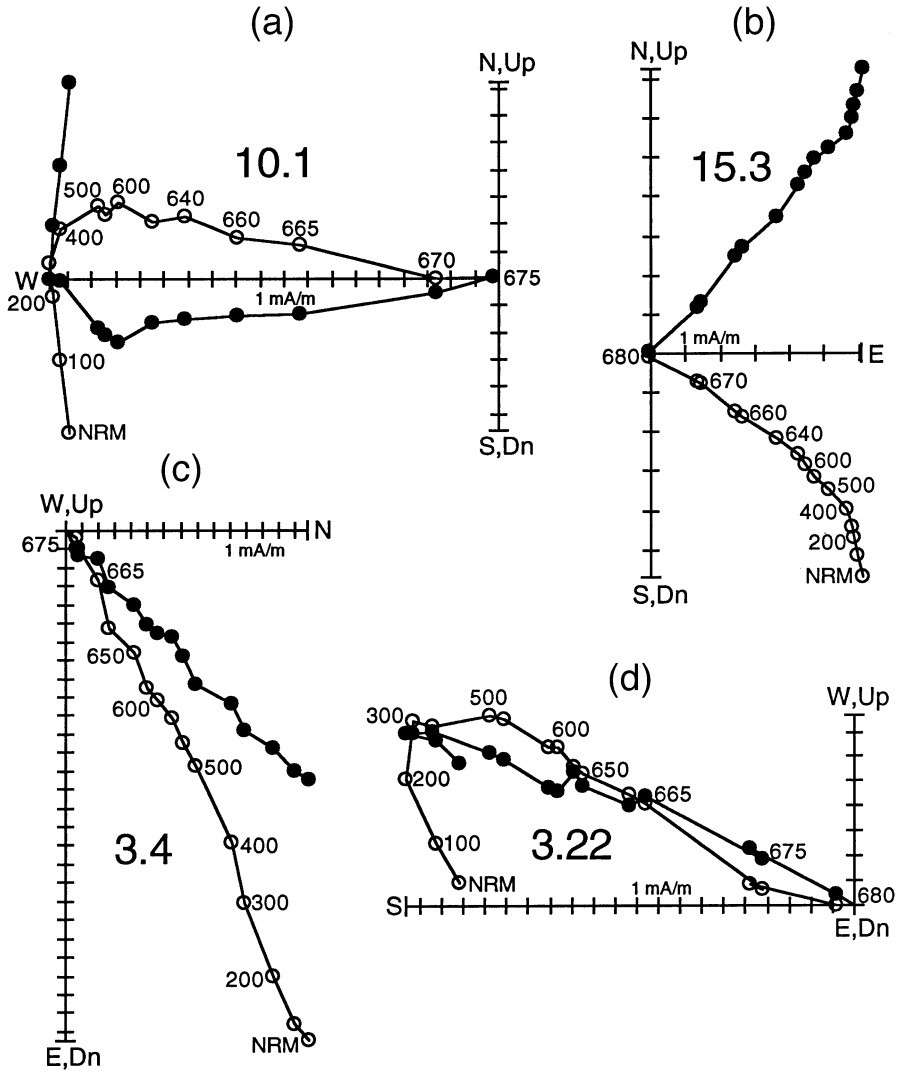


Figure 4. Representative orthogonal plots for the samples from Sangzhi. (a) and (b) are from Member 2 on opposing limbs of the fold, and (c) and (d) are from Member 3. Symbols and coordinates as in Fig. 2.

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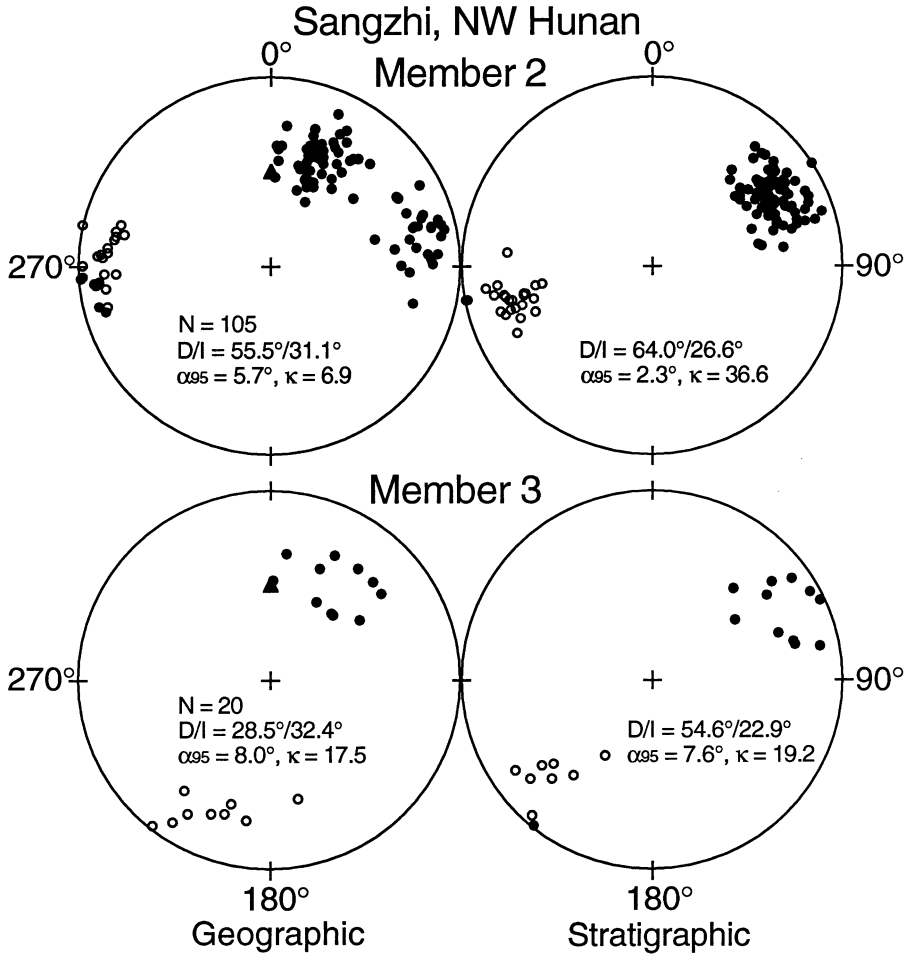


Figure 5. Equal-area projection of the ChRM directions of the samples from Sangzhi. Symbols as in Fig. 3.

a normal-polarity magnetozone. Farther up-section, the remaining samples from Members 3 and 4 are all of reversed polarity (Fig. 8).

Nanzhang, central Hubei

Member 4 was also sampled at the Liujiatai section on the northern flank of the synclinal Jingmen–Dangyang Basin (Fig. 1). 61 samples, including two from the basal part of Member 5, were collected over about 410 m of the section. In addition, 10 sites from the Liujiatai section and 11 sites from a second (Maoping) section on the western flank of the basin were drilled from Member 2.

The palaeomagnetic results for Member 2 have been reported in Huang & Opdyke (1997). In summary, the NRM of most samples from Member 2 consists of two components: a low-temperature component (A) probably acquired at a late stage of folding; and a high-temperature component (B) which passes a fold test and is of dual polarity (Figs 6a, 6b and 7; see also Figs 2 and 3 in Huang & Opdyke 1997). The demagnetization behaviour of most samples from Member 4 is similar to that of Member 2; 46 samples, including the two from Member 5, yielded well-defined high-temperature dual-polarity ChRM directions (Figs 6c, 6d and 7). The relative sample VGP latitudes define four magnetozones from bottom to top of Member 4: R–N–R–N (Fig. 8). The reversed magnetozone at the bottom is consistent with that observed in the two samples taken from the basal part of Member 4 at Sangzhi, and apparently continues with the upper reversed zone in Member 3 observed there. The top normal magnetozone extends into Member 5, but its upper limit is undetermined.

DISCUSSION

As we have demonstrated above and elsewhere (Huang & Opdyke 1996, 1997), the ChRM resolved from Member 2 at Badong, Sangzhi and Nanzhang is pre-folding with dual polarity. The ChRM resolved from Member 3 at Sangzhi and Member 4 at Nanzhang is similar to that of Member 2 from the corresponding sections in direction and unblocking temperature spectra. The primary nature of the ChRM from these rocks is further supported by the fact that the same magnetic polarity zone has been observed in comparable stratigraphic levels at more than one section, for example the reversed polarity magnetozones across the boundaries between Members 2 and 3 and between Members 3 and 4. This leads us to believe that the polarity pattern so determined recorded the polarity change of the geomagnetic field when the sediments were deposited or shortly afterwards.

A composite magnetic polarity sequence for the Badong Formation is constructed using the data described above (Fig. 8). Nine magnetozones are recognized, with three longer

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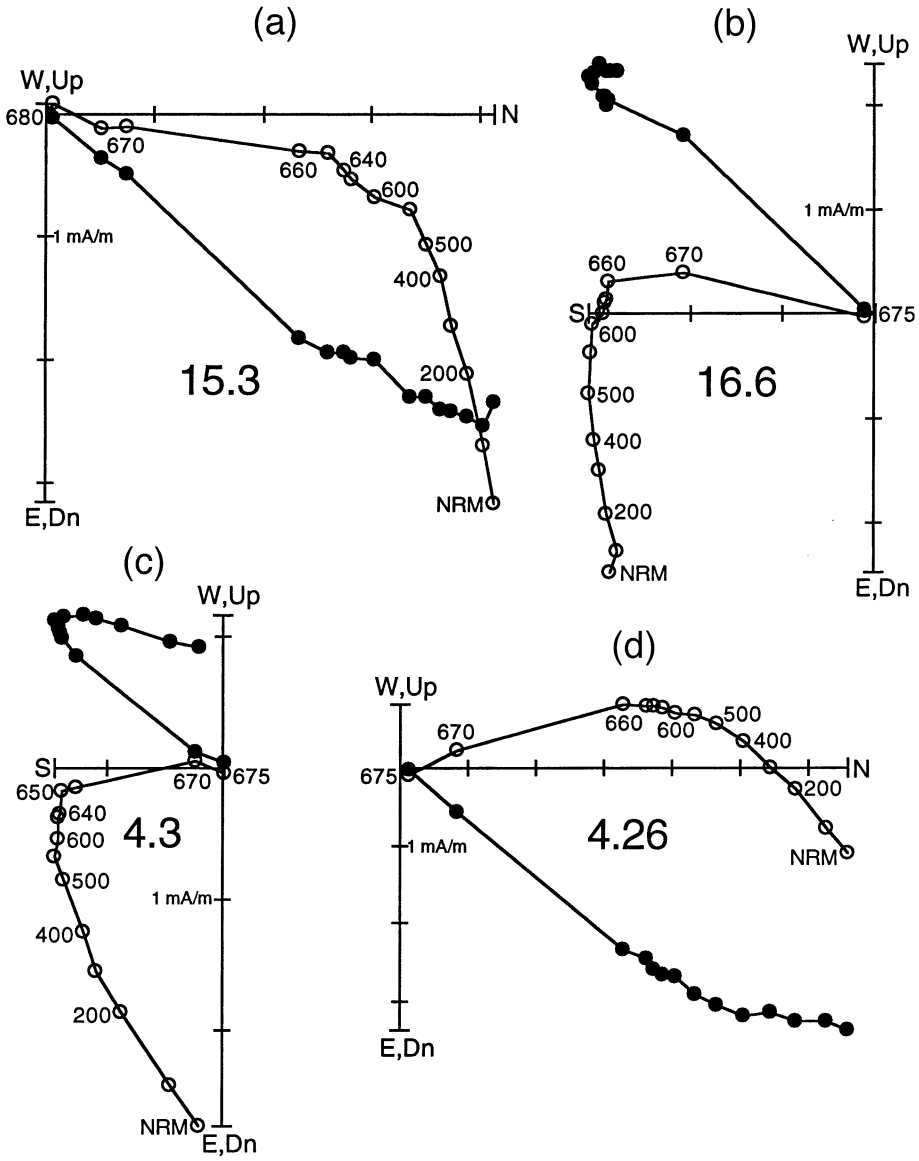


Figure 6. Representative orthogonal plots for the samples from Nanzhang. (a) and (b) are from Member 2 on different flanks of the fold, and (c) and (d) are from Member 4. Symbols and coordinates as in Fig. 2.

zones (N2 to N3, R3, and N4) dominating the sequence. A possible short normal magnetozone within R1 is indicated by a single normal-polarity sample. Members 1 and 5 lack reliable data and the magnetozones that they may contain are undetermined. It should be pointed out that an underlying assumption of equal sedimentation rate for all the members is adopted for the compilation. This may not hold true for Member 3, which could have a lower sedimentation rate than Members 2 and 4. The difficulty is, however, that it is hard to know how different the sedimentation rates might be between Members 3 and 2 or 4. Nevertheless, since Member 3 contains basically a pair of normal and reversed magnetozones, even if their duration is increased it will not alter the correlation with the magnetic polarity sequence reported from the western Tethyan region, as shown below.

Magnetostratigraphic data for the Middle Triassic pelagic carbonates have recently become available from the western Tethyan region. For purposes of comparison, a synthetic Mid-Triassic MPTS (Fig. 8) is compiled primarily from the data reported by Gallet *et al.* (1998) and Muttoni *et al.* (1996, 1997, 1998). In the compilation, the lower portion of the Anisian Stage (Kc1r to Kc3n) is based on Muttoni *et al.*'s (1996) study on the Kcira section in Albania. The overlying reversed magnetozone N1r was recently determined by Muttoni *et al.* (1998) at the Nderlysaj section in the Albanian Alps. Currently, there is a data gap in the upper Illyrian Substage. Muttoni & Kent (1994) observed an Illyrian interval of normal polarity from the Prezzo Limestone in the western Southern Alps, but later suspected it to be a remagnetization on the grounds that the (ammonoid) biostratigraphy seems to correlate the Prezzo Limestone largely with the Dont-Monte Rite section in the Dolomites region of Italy, which yields a reversed-polarity interval correlative with N1r determined from the Nderlysaj section (Muttoni *et al.* 1998). Still, it is possible that a normal-polarity interval of relatively long duration might exist in the Illyrian Substage. This is because normal magnetic polarity

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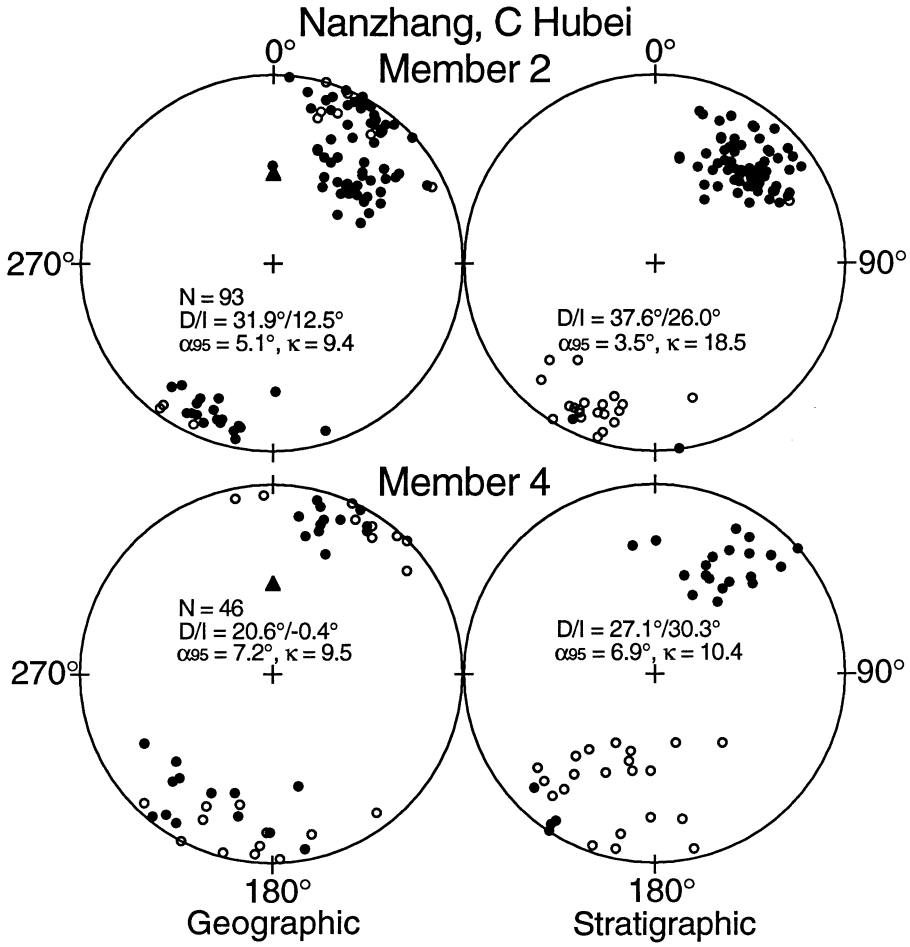


Figure 7. Equal-area projection of the ChRM directions of the samples from Nanzhang. Symbols as in Fig. 3.

has been detected immediately below (N2n) and above (F1n.1n) the data gap, and Gallet *et al.* (1998) also observed a normal-polarity interval in a condensed section (Gamsstein 2) of Austria which spans the Upper Anisian to Ladinian. Hopefully, future magnetostratigraphic investigations on longer Mid-Triassic stratigraphic sections will help to resolve such uncertainties. The polarity sequence for the Ladinian Stage follows Gallet *et al.* (1998), but equal sedimentation rate, instead of equal biozone duration, is assumed. The reversal frequency during the Anisian (1.8) seems to be lower than that during the Ladinian (2.8 reversals per million years); more importantly, the Anisian Stage appears to be dominated by major magnetozones (Kc2n to Kc3n and N1r), whereas the Ladinian Stage is characterized by frequent reversals of shorter duration.

Comparing the polarity sequence of the Badong Formation with the Mid-Triassic MPTS derived from the western Tethyan region, the long reversed magnetozone, R3, which straddles Members 3 and 4 of the Badong Formation, appears to correlate best with N1r observed in the western Tethys. Consequently, the overlying magnetozones N4 to N5 of the Badong Formation would correlate with either N2n to A+ or N2n to C+ in the western Tethys, depending upon whether a normal-polarity interval of longer duration exists in the upper Illyrian Substage. This correlation is consistent with the conodont fossil data. The conodont data from the Kcira and Nderlysal sections indicate that the *Neospathodus kockeli* zone basically spans the Pelsonian Substage; in terms of magnetostratigraphy, it encompasses the upper portion of Kc3n and a major portion of N1r (Muttoni *et al.* 1996, 1998). As described earlier, Member 3 of the Badong Formation yields a similar conodont zone. Thus, a conclusion can be drawn that the bulk of the Badong Formation is Anisian in age and the Ladinian sediments are largely missing in the study area.

This conclusion is in good agreement with palaeontological and stratigraphic evidence from South China. Yin (1982) pointed out that the controversy over the Ladinian Stage in China centres on two index fossils, *Asoella illyrica* and *Progonoceratites*. He believes that distinct features suggest that previously identified bivalve *Asoella illyrica* is actually *Leptochondria illyrica*, which thrived in the Anisian and rarely extended into the Ladinian, and that the appearance of *Leptochondria illyrica* and associated bivalves marks the lower boundary of the Asinian in China. This view has been accepted by most Chinese palaeontologists (Fang, personal communication). As to *Progonoceratites*, they often occur with some other Anisian ammonoids such as *Beyrichites*, *Noetilingites*, and *Leptochondria* in the same horizon in China and it is thus better to consider them to be Anisian in age (Yin 1982). Furthermore, regional stratigraphic correlation suggests that the YB and the Qinling Fold Belt (QFB) experienced a major, maybe the final, regression during the Ladinian which resulted in partial or complete lack of sedimentation, or in some places terrestrial deposits. The Ladinian regression in these areas

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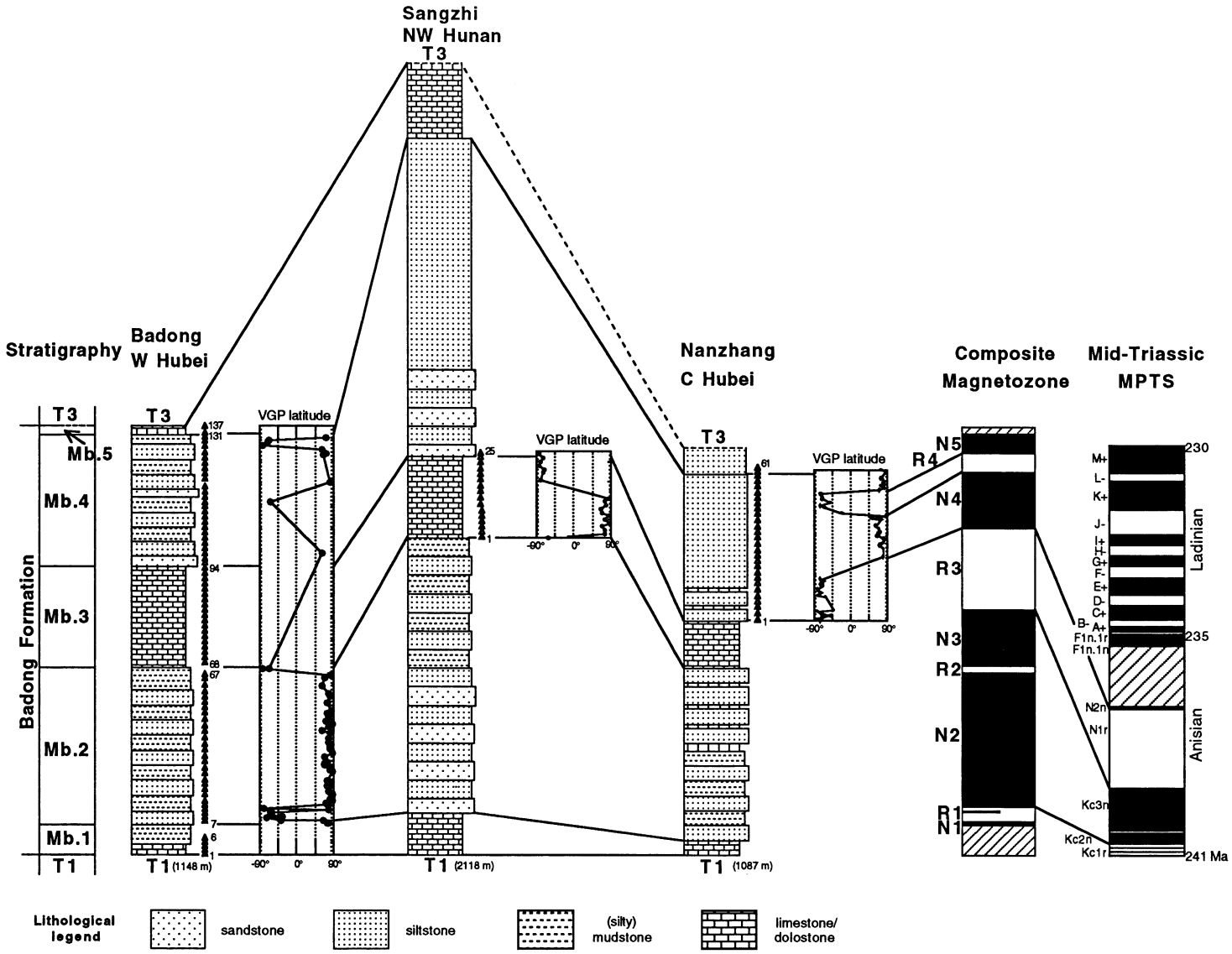


Figure 8. Lithology and polarity sequence of the Badong Formation observed at three studied sections as compared with the Mid-Triassic MPTS derived from the western Tethyan region. The small triangles along the lithological columns are sample positions, and the numbers next to the triangles are sample numbers. The thickness of the Badong Formation at each section is indicated under the lithological column. Sample VGP latitudes were calculated with respect to the overall mean (north) palaeomagnetic pole for each section. The composite magnetozone for the Badong Formation was constructed based on the stratigraphic division and thickness of the type section at Badong. Filled (open) bars represent normal (reversed) polarity zones; half-bars show single-sample results. The Mid-Triassic MPTS was compiled from the data reported by Gallet *et al.* (1998) and Muttoni *et al.* (1996, 1997, 1998). Names of the magnetozones given by the original authors (except L1n) are indicated to the left of the column, and the absolute ages for the Mid-Triassic Stage boundaries indicated to the right of the column are from Australian Geological Survey Organization (1995).

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signals the amalgamation of the YB with the NCB (Yin 1982; Chen *et al.* 1998). Lack of Ladinian sediments also explains the long-term puzzling fact that the bivalve assemblage *Halobia Daonella* which is commonly seen in the Ladinian rocks in south-western China has not yet been found in the Badong Formation and equivalent rock formations of the YB (Zhang *et al.* 1987).

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